

The Alberta Buck — Privacy

A payment between citizens, seen from the middle

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A citizen should be able to prove who they are without wearing a name tag on every purchase. A counterparty should be able to keep an honest receipt. A court should be able to examine evidence. These are compatible ambitions.

Bob wants to pay Carol 100 BUCK. Mallory wants to watch. We will build the payment one layer at a time, beginning before either citizen has an account.

Ethereum is an open ledger. Its addresses, transactions and token movements are observable. The privacy question is which observations can be joined: an account to a person, two accounts to one owner, or a payment into a pool to a payment out. Like a VPN or onion router, Notes introduce separation between endpoints. Here the separation comes from funded commitments and proofs, rather than a chain of packet relays. The result sought is *pseudonymity and compartmentalisation*: a public business account, private savings and everyday spending need not advertise their common owner or all their relationships. Encryption is a door. Privacy also needs walls.

This paper describes the current v2 design and prototype, not an operating Alberta government service. Its credential privacy argument awaits independent review, and its development proof setups require production ceremonies before deployment. The illustrations assume honest certification, sound cryptography and wallets that retain their secrets.¹

How to read the pictures

Bob stays left, Carol right, Mallory in the middle. The lower strip names the *separate* issuer, wallet checker or blockchain contract doing the work; Mallory is never the verifier merely because she occupies the middle column.

Each picture carries forward the relevant contents of the last. A *plus* adds material, a *lock* marks encrypted contents, an *opening arrow* exposes contents to their intended reader, and a *check* marks a completed check. An arrow crossing Mallory's column gives her a view of the traffic, not a decryption key. Her column says what she can actually read or do.

We need only a few objects:

- A **record** contains the name and other certified particulars. Its fixed encoding produces an identity fingerprint, called **M** in the pictures. A fingerprint lets a wallet check a record; it does not spell out a name.
- A **credential** is the issuer's signature on that identity. Its private original stays in the wallet; a **presentation** proves possession without publishing the original.

- An **envelope** is encrypted data. A **proof** lets a checker verify a particular fact about its contents without opening it.
- A **commitment** is a sealed description with a short public fingerprint. Its **opening** supplies the hidden contents and randomness. A **spent marker** records redemption once, without normally identifying the commitment.
- A **membership root** is a published summary of certified records. A **path** connects one record to that summary; a private membership proof can establish the connection while hiding the record and path.

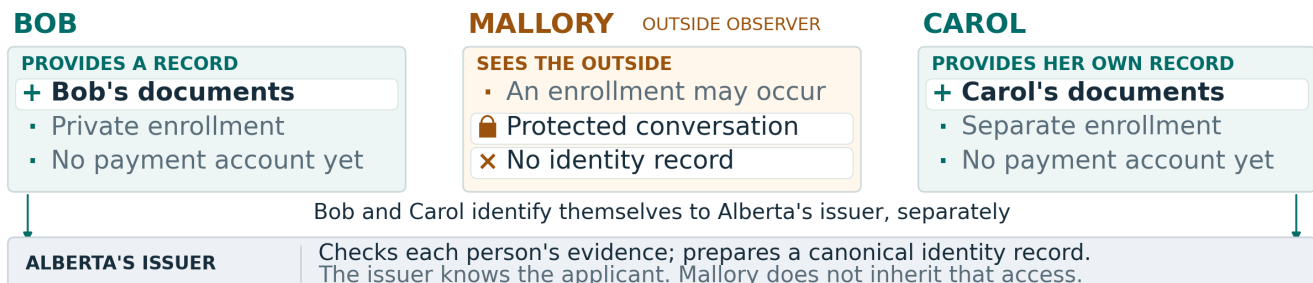
The three Notes stories are alternatives for the same payment, not conversions of B1 into A1 into A2. We give the 100-BUCK note a place in a 500-BUCK batch so that minting does not announce its individual value. Earlier disclosures are never erased by changing note style.

First, give the person a credential

I1. Alberta's issuer meets Bob; separately, it meets Carol

Each applicant provides the evidence the issuer requires. The issuer checks it and prepares a consistently encoded identity record. Bob and the issuer know Bob's record. Carol's enrollment is a separate conversation.

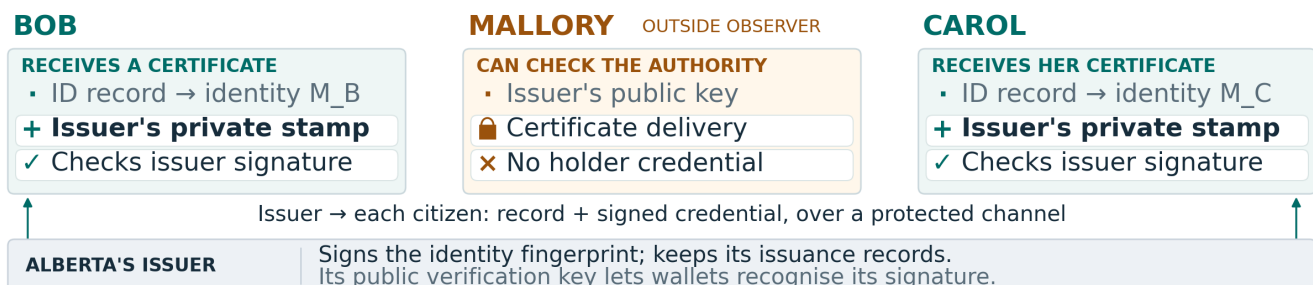
Mallory may notice an appointment or a network connection. A protected enrollment channel keeps the submitted documents out of that observation. Cryptography preserves the issuer's statement; the quality of the original identity check remains the issuer's responsibility.



I2. The issuer returns a signed credential

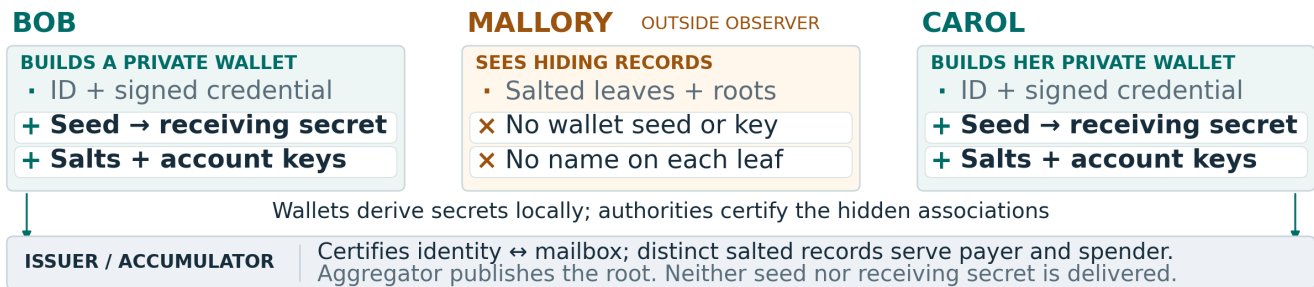
The issuer fingerprints the canonical record and signs it. Each citizen receives the record and credential through protected delivery, and checks the signature against the issuer's recognised public key. The issuer keeps its own issuance evidence. The raw credential is not published on Ethereum.

This is certification, not an account password. The issuer already knows the identity it certified. The objective is to avoid giving it a new test for recognising every future account carrying that identity.



I3. The citizen makes the keys

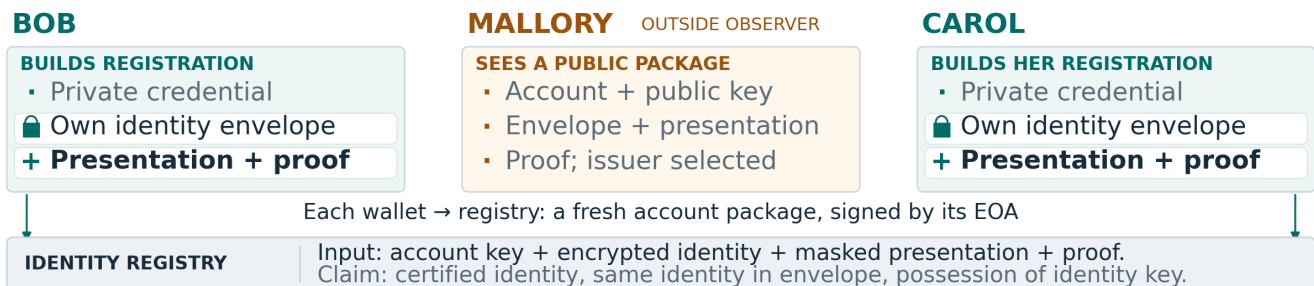
At enrollment, each wallet derives an independent **receiving secret** from its owner's secret seed, using a dedicated key-derivation domain. Only the public receiving key is available to payers. The seed restores the private key; an identity record cannot. A photocopy of your identification should not open your mail. Enrollment also certifies the identity-to-mailbox association in salted commitments. One supports later private spending proofs; a separate one lets a payer check the public receiving key. Different salts keep the two records from being trivially matched. The seed and receiving secret stay with the citizen. Authorities know the associations they certify; public observers see hiding records and their published root.²



I4. The wallet packages a fresh account

An externally owned account, or **EOA**, has an Ethereum signing key. BUCK adds a separate identity encryption key for that account. Neither is the Notes receiving secret.³

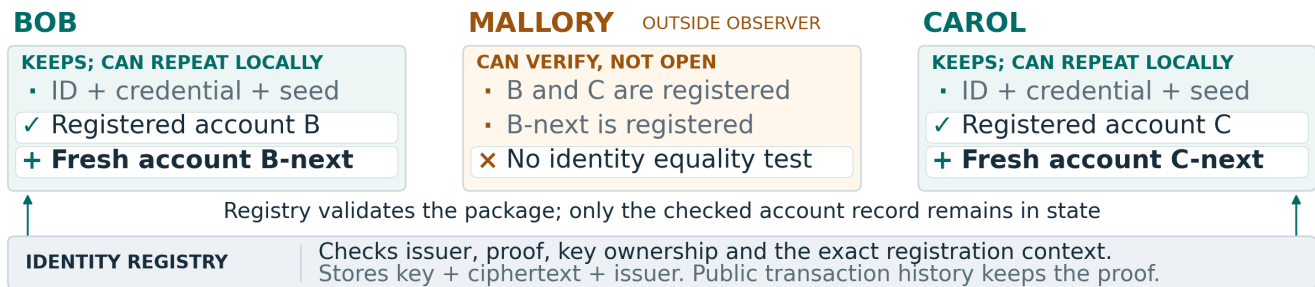
Bob's wallet encrypts his identity fingerprint under his new identity key. It adds a freshly masked credential presentation and a proof: *this is a certified identity, this envelope contains that identity, and I hold the corresponding identity key*. Bob's EOA signs the registration transaction. Carol does the same. The credential's original signature stays private.



I5. The registry checks the package, without opening it

The registry checks the issuer, proof and key ownership, including the specific account, chain, registry and action to which the proof belongs. It stores the account's public identity key, ciphertext and issuer. The transaction history retains the presentation and proof.

Bob can repeat this locally with fresh keys and randomness. The design's credential presentation gives Mallory no identity-equality test between accounts, even if she later learns Bob's record. Funding links, reused addresses and network observations are separate clues. Fresh cryptography does not make an old public connection disappear.



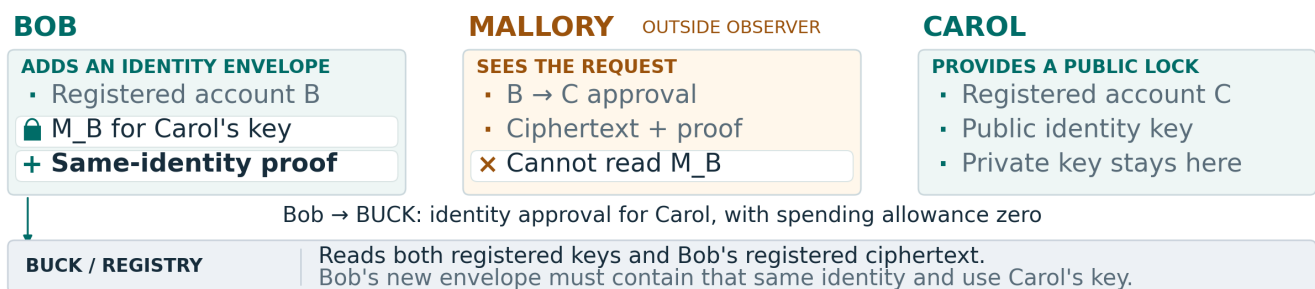
Now, pay between two EOA accounts

Here Bob and Carol use private-identity accounts. They first establish identity evidence in both directions, then Bob authorises the payment. The chain will see the payment between their addresses. The envelopes protect the identities inside that visible relationship.

E1. Bob addresses an identity envelope to Carol

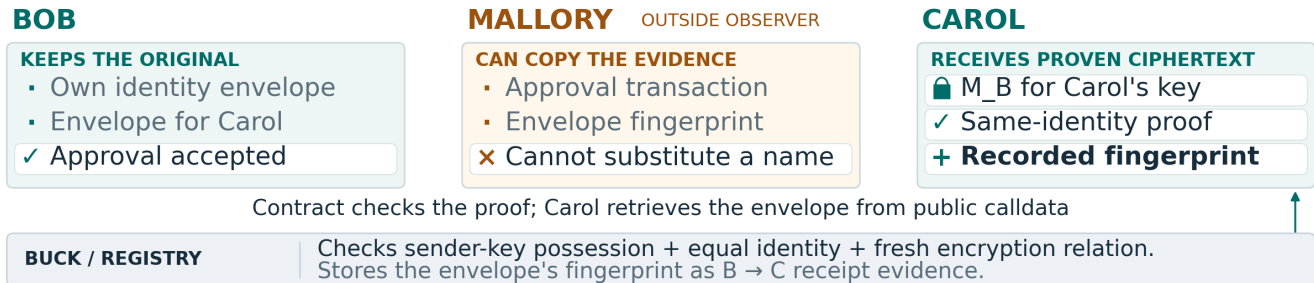
Bob reads Carol's registered public identity key and makes a fresh encryption of his own fingerprint for her. He adds a proof that its contents match his registered identity, and calls BUCK's identity-bearing approve.

For this direct **transfer** example, the spending allowance is **zero**. This creates the identity evidence without giving Carol permission to withdraw Bob's funds. A nonzero allowance would additionally authorise delegated spending up to that amount. Identity disclosure and authority to move money are distinct jobs, even when an API can do both.



E2. The contract verifies the envelope's provenance

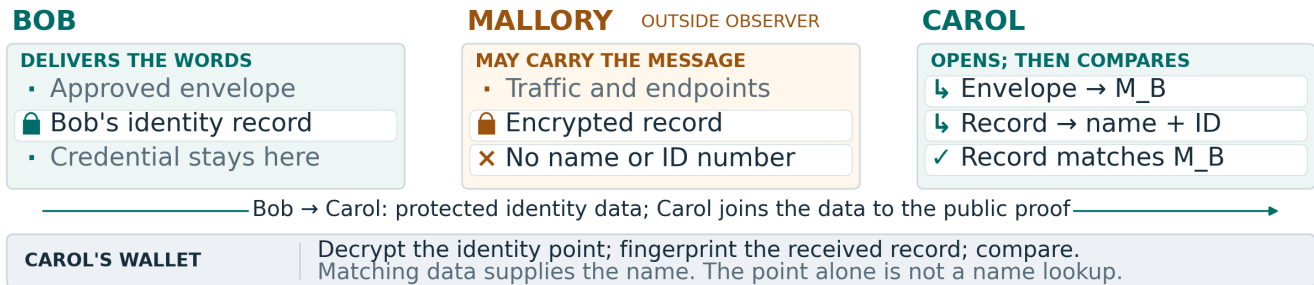
The checker uses Bob's existing registry record, not a replacement supplied by the request. It verifies possession of Bob's identity key and that the new encryption carries the same identity under Carol's key. BUCK records a fingerprint of the checked envelope. The envelope and proof remain available in the transaction's public input data. Mallory can copy them, but cannot replace Bob's identity with another person's and preserve the proof. Carol now has something worth opening.



E3. Carol opens the envelope and checks the name

Carol decrypts with her account's private identity key. Out comes Bob's fingerprint, M_B . Bob separately delivers his canonical identity record over an authenticated, encrypted data channel. Carol fingerprints that record and compares it with the decrypted point.

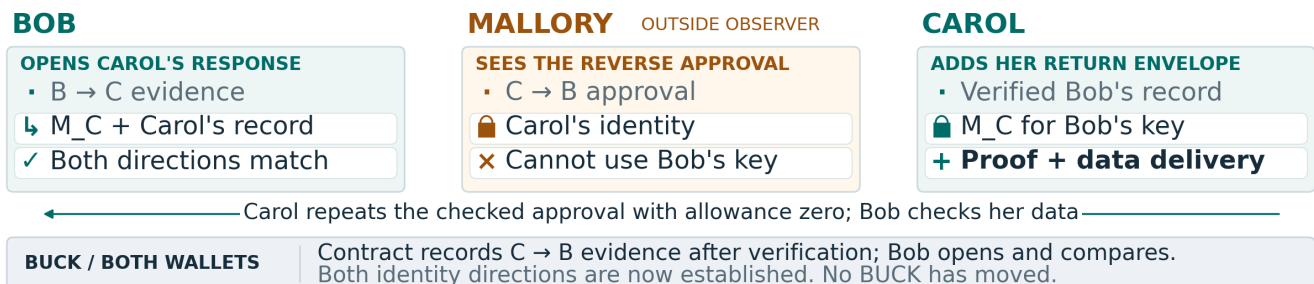
The proof answers *is this the identity bound to Bob's account?* The record answers *whose identity is it?* Both must agree. Ethereum cannot itself ensure that a human-readable record was delivered. The wallet checks delivery before accepting the relationship. Holochain is the companion papers' proposed data channel; the diagram does not assume a deployed hApp.



E4. Carol returns the same assurance

Carol encrypts her identity for Bob, supplies the corresponding proof and establishes the reverse receipt fragment, again with allowance zero. She sends her record privately; Bob opens the envelope and checks the match.

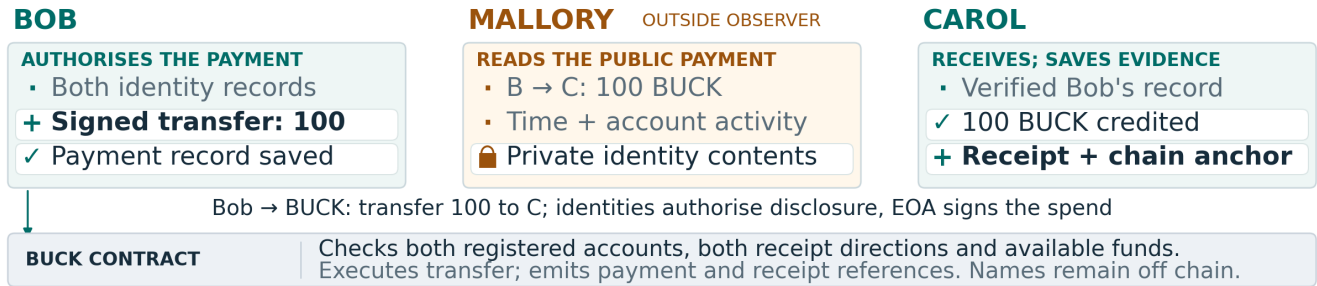
The construction now contains two checked directions. Each citizen has received a certified identity record. Mallory has observed two approvals between addresses. No payment has happened yet. Recognising someone is not the same as paying them.



E5. Bob signs the payment; the contract executes it

Bob's EOA signs `transfer(Carol, 100)`. BUCK checks the registered endpoints, both identity directions and available funds, then executes the transfer and emits its receipt references. The wallets retain the identity records, proofs and confirmed payment reference together.

Mallory sees the two addresses, 100 BUCK and the time. Learning who owns either address elsewhere can identify this activity. The gain here is confidential identity binding. The next gain is separating the payment's two ends.⁴

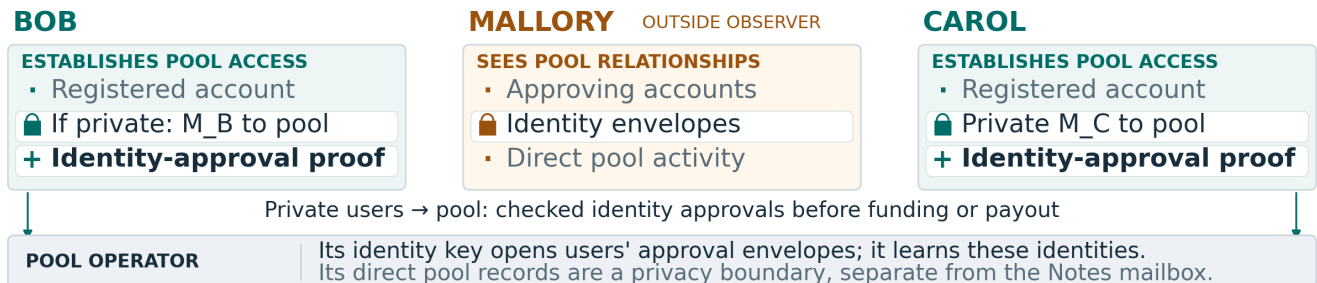


The Notes pool is also a participant

N1. Establish the pool relationship

Notes hold BUCK in a contract and pay it out when the required proofs pass. The BUCK transfers into and out of that pool retain the ordinary identity rules. A private Bob identifies himself to the pool before funding it; private Carol does so before receiving its payout. They use the same envelope-and-proof procedure just illustrated.

The pool operator holds the identity key bound to the pool and can decrypt these approvals. It can associate identities with direct pool activity. This is an institutional privacy boundary, not an invisible intermediary. A1 and A2 still avoid publishing an exact note-to-redemption link; the operator's records and timing clues must be considered separately.⁵



B1: a bearer note from a public issuer

Bob writes the note; he is now the *note issuer*. Alberta remains the *identity issuer*. These are different roles.

For B1 and A1, Bob uses a publicly identified note-issuing contract under his control: his “note desk.” This matches the current public-issuer contract-binding interface; ordinary EOA registration does not itself switch on a public-identity flag.⁶

B1.1. Bob prepares a bearer opening

Bob chooses the 100-BUCK face value and a fresh random opening secret. He packages the note's B1 style, value and public-issuer material into a commitment. His registered identity key authenticates the batch. There is no recipient lock yet. Whoever obtains the complete bearer opening can seek redemption, provided they meet the identity requirements. Bob's published identity lets a future holder know who wrote it.

BOB

BUILDS A BEARER NOTE

- + **Public issuer record**
- + **100 + secret opening**
 - No chosen recipient

MALLORY OUTSIDE OBSERVER

CAN IDENTIFY THE ISSUER

- Bob's public note desk
- Published identity
- ✗ No opening yet

CAROL

CAN BE ANY FUTURE HOLDER

- Own registered account
- Own private identity
- No mailbox supplied

Bob prepares a B1 note: value + random opening + public issuer material

BOB'S PUBLIC NOTE DESK

A contract authorised by Bob and bound to his certified identity key.
Draft note → hiding commitment; signature authenticates the public issuer.

B1.2. Funding turns the draft into a note

Bob's desk submits the commitments, issuer signature and batch-mint proof. The proof establishes that the committed values add to 500 BUCK and that the commitments extend the current note tree correctly. The contract checks the public-issuer mode and signature, and pulls the funding.

Funding and insertion succeed together or revert together. Mallory sees the issuer, total, batch size and commitments. She does not receive the individual openings. A convincing picture of a note would not establish any of these facts.

BOB

FUNDS A BATCH

- B1 opening for 100
- + **Batch proof + signature**
- ✓ 500 BUCK escrowed

MALLORY OUTSIDE OBSERVER

READS THE MINT RECORD

- Public issuer + 500 in
- Batch size + commitments
- ✗ No per-note openings

CAROL

CAN VERIFY THE FUNDING

- Own wallet
- + **Public mint reference**
- No payment received yet

Bob's desk → Notes: funded batch containing this 100-BUCK bearer note

NOTES / BUCK

Checks mint proof, issuer signature, batch mode and current tree position.
Moves 500 into escrow and inserts the commitments in one atomic transaction.

B1.3. Carol receives and inspects the bearer package

Bob hands Carol the opening and mint reference, physically or through protected delivery. Her wallet reconstructs the commitment, authenticates the mint and issuer, and checks whether its spent marker has appeared.

Here interception is theft-enabling: Mallory with a copied opening and an eligible account can race Carol. Bob can retain a copy too. The check says *unspent now*, not *reserved for Carol*. Bearer delivery is convenient precisely because possession carries authority; copies inherit it.

BOB

HANDS OVER THE NOTE

- Funded commitment
- + **Opening + mint ref**
- Can retain a copy

MALLORY OUTSIDE OBSERVER

INTERCEPTION MATTERS

- ✗ Copied opening = bearer
- Can race if eligible
- Courier must protect it

CAROL

UNPACKS THE BEARER NOTE

- ↳ 100 + secret opening
- ✓ Commitment + mint match
- ✓ Not spent as of check

Bob → Carol: bearer opening by protected message or physical handover

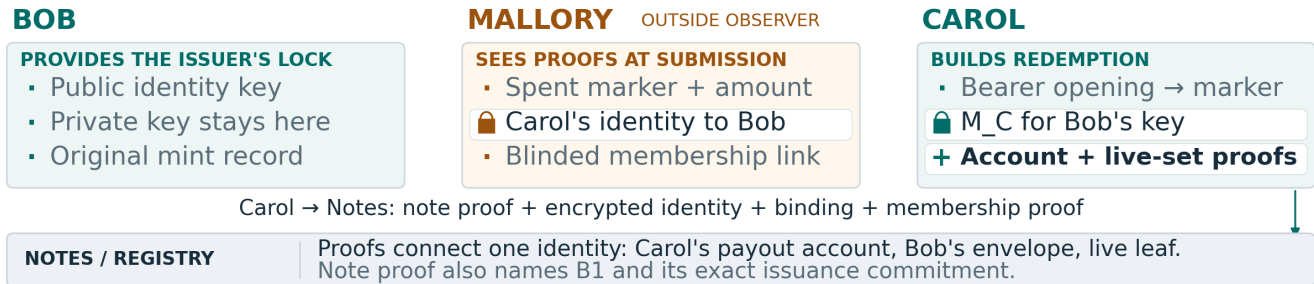
CAROL'S WALLET

Rebuilds commitment; checks issuer, funding, membership and spent status.
A copy is spendable too. Validation is not a reservation of the money.

B1.4. Carol adds an envelope identifying the depositor

Carol derives the spent marker from her opening and prepares a note-spend proof. She also encrypts her own identity for Bob's registered issuer key. A binding proof ties that envelope to the identity in her payout account; a membership proof establishes current certification of that same identity.

Those checks share a properly blinded identity commitment. Mallory cannot substitute a convenient certified person in one check and a different person in the other. Carol's separate approval to the pool from N1 is also in place before payout.



B1.5. The contract checks the layers and pays

Notes verifies that this is a funded *B1* note, that its marker is unused, that the named issuer is the original issuer, and that Carol's identity binding and sufficiently recent membership proof pass. It records the marker and pays 100 BUCK atomically. A second redemption fails.

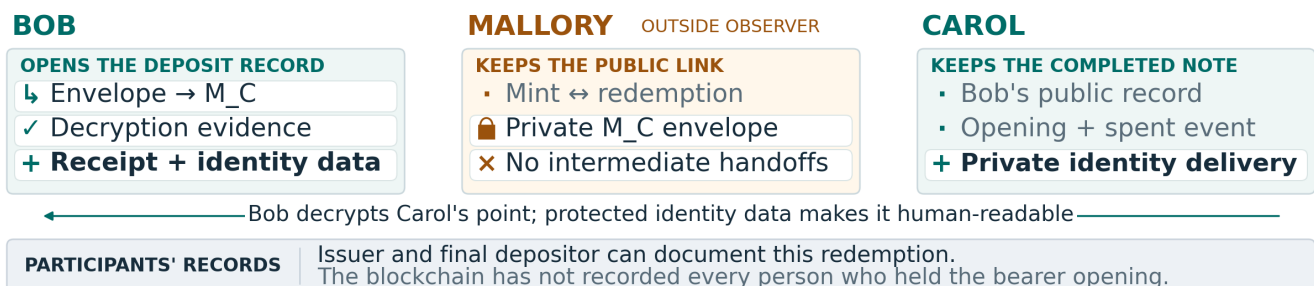
The current B1 interface supplies the **exact issuance commitment** publicly to authenticate the issuer. Mallory can therefore link this redemption to its mint. Carol's private identity remains encrypted, but the mint-to-payout edge does not. This distinction is essential to the comparison.⁷



B1.6. Bob opens the depositor's envelope

Bob reads the spend event and decrypts Carol's identity point. A verifiable decryption proof lets him substantiate what it contains without handing over his private key. A separately delivered canonical record, or an authorised match against issuer records, supplies the human name.

Until that record is available, he has a certified point, not a complete named receipt. The evidence concerns Bob and the final depositor. B1 may have passed through other hands; those handoffs are not magically written into its blockchain history.

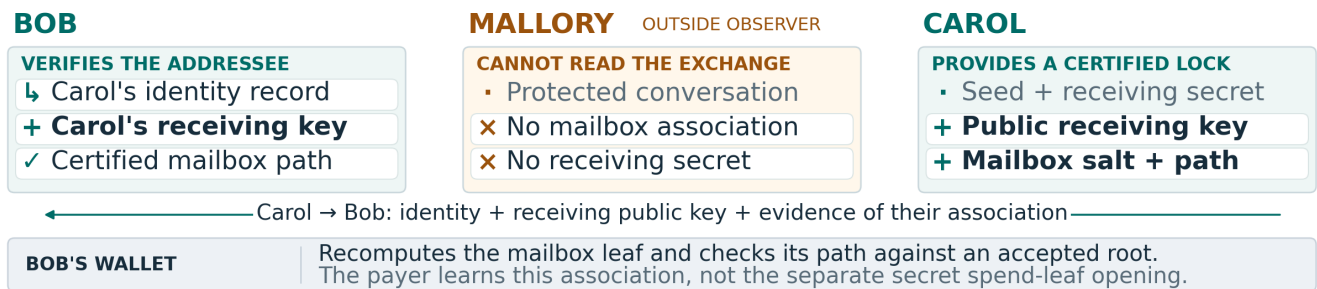


A1: address the note to Carol

A1 adds a recipient. Bob remains publicly identified, but a copied note package is no longer a bearer authorisation. Carol can redeem through a registered account of her identity without giving Bob its address in advance. Her mailbox and her payout account do different jobs.

A1.1. Carol supplies a certified public receiving key

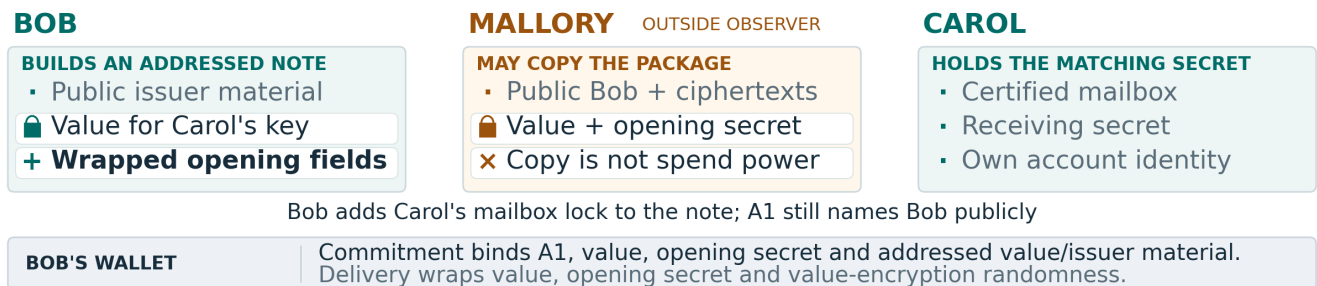
Carol sends Bob her identity record, public receiving key and evidence of their certified association. Bob checks the mailbox commitment's salt and path against an authenticated, acceptable root before paying. This is a private disclosure to Bob. It lets him recognise that mailbox association, so it belongs in his confidential records. Its salt differs from the hidden spending association's salt. Neither the receiving secret nor Carol's seed travels. A public key is the lock she lends him, not the key that opens it.



A1.2. Bob adds Carol's lock

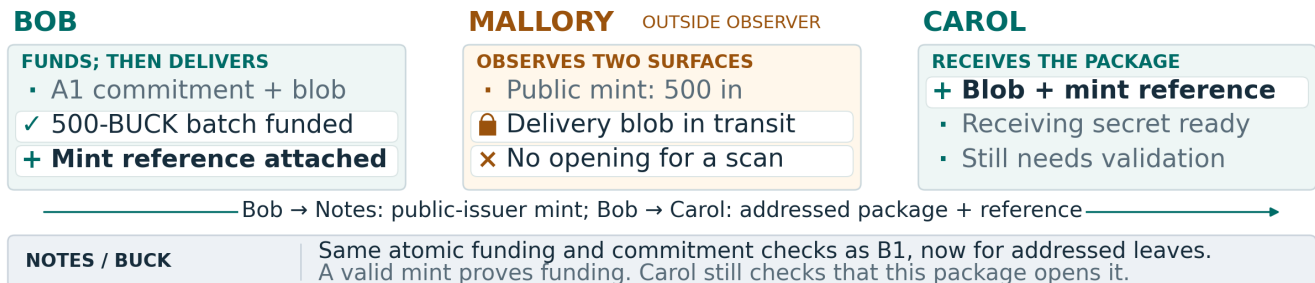
Bob encrypts the value for Carol's receiving key. The note's committed identity payload combines that encrypted value with his public issuer material. The delivery also carries an encryption of Carol's identity for her own mailbox.

The opening secret, value and value-encryption randomness travel as separately masked fields in the delivery blob. Bob can construct the masks from his fresh encryption randomness and Carol's public key. Carol can reconstruct them with her private receiving secret. A courier can do neither.



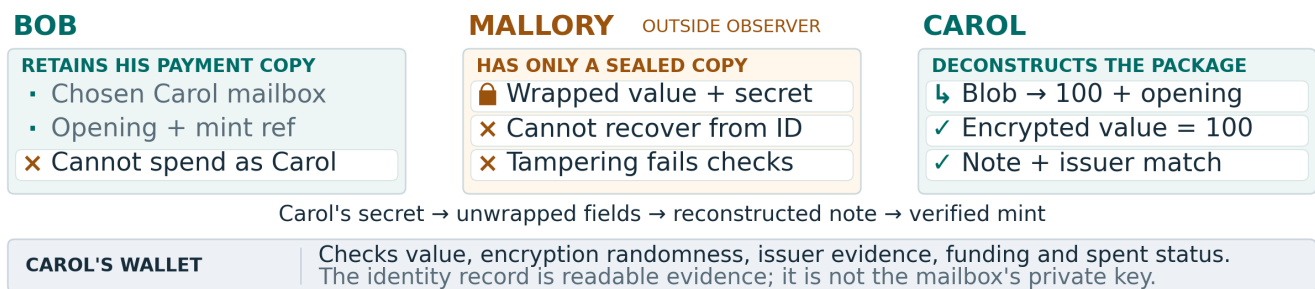
A1.3. Mint, then deliver the funded package

The same batch funding checks run, this time for addressed notes from a public issuer. Once the mint is confirmed, Bob supplies Carol with the delivery blob, issuer evidence and mint reference. Mallory sees 500 BUCK enter the pool and may intercept the blob. The wrapped opening prevents her from computing this note's spent marker or reading its value from that package. Endpoint and traffic observations remain possible. Transport privacy still has work to do.



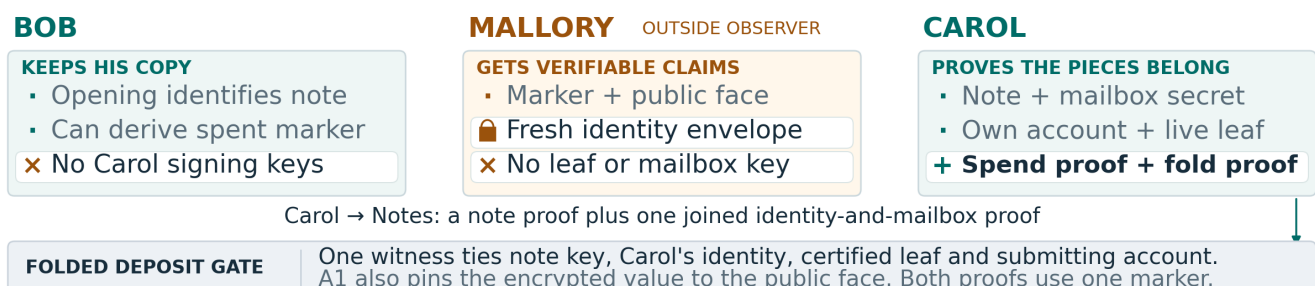
A1.4. Carol takes the package apart

Her receiving secret opens the masked fields. Her wallet checks that the value ciphertext contains 100, that the supplied randomness fits, and that the recovered opening reconstructs a commitment in the authenticated mint. It checks Bob's public-issuer evidence and the spent status too. Each wrapper has now served its purpose: delivery protection becomes private witness material in Carol's wallet. A wrong key or altered field fails the relevant consistency checks. Bob's retained opening still cannot satisfy Carol's identity-and-mailbox spending requirement.



A1.5. Prove that the pieces belong to one citizen

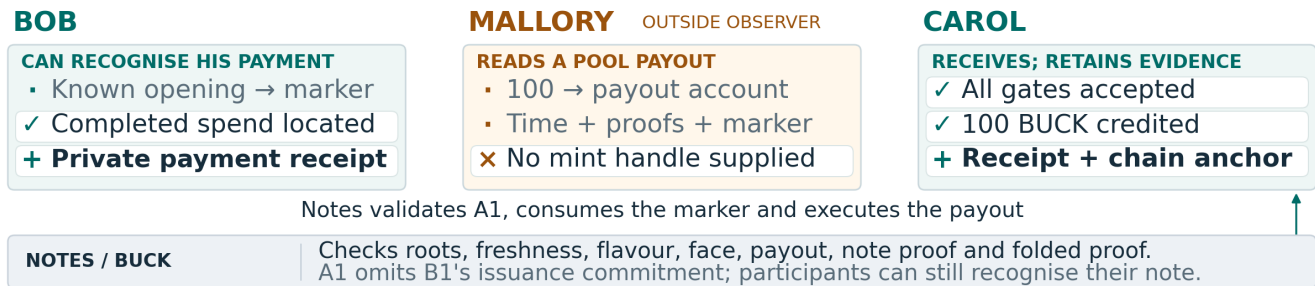
Carol chooses her submitting account and prepares two proofs: one for the funded note and one *folded* proof joining the identity conditions. The latter checks that her receiving secret opens this note, her account holds the matching identity, and a certified hidden leaf binds that identity to that receiving secret. A1 also checks the encrypted value against the public payout amount. The proofs share the note's spent marker. Showing "I have a key" beside "I have an identity" would not be enough. The checked association makes them belong to the same citizen.⁸



A1.6. Redeem without publishing the mint handle

Notes checks both proofs, their roots, the A1 style, face, payout address and unused marker. BUCK's pool-to-recipient identity checks also apply. The operation consumes the marker and pays 100; any failure reverts the state changes.

A1 supplies no exact issuance commitment at spend. Mallory sees the payout account, amount and time, with no explicit mint handle to join. Bob and Carol can still derive the marker from their retained opening and recognise this payment. Privacy from a stranger need not mean amnesia for the participants.



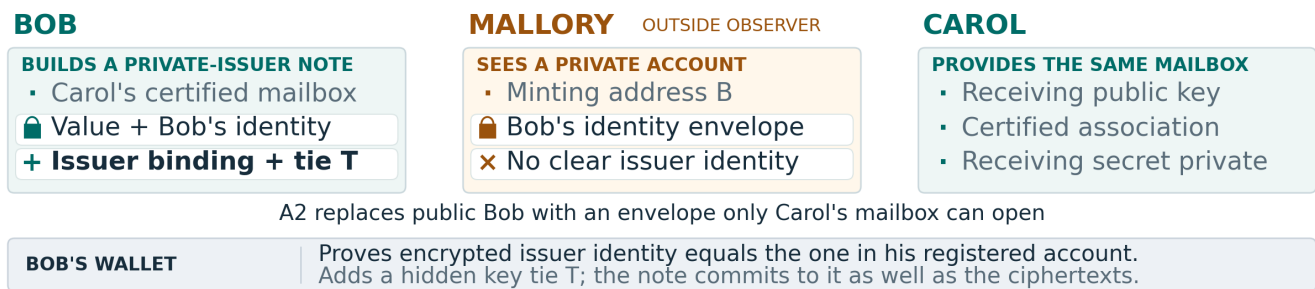
A2: put Bob's identity inside the envelope too

A2 keeps the addressed protection and makes the note issuer's identity private. Read this as an alternative using an appropriately separate private account. Publishing Bob's name in the B1 story did not become reversible when we turned the page.

A2.1. Add a private issuer envelope and a key tie

Bob uses Carol's already verified mailbox key. Alongside the encrypted value, he encrypts *his own identity* for that mailbox. An issuer-binding proof connects this envelope to his registered account while hiding the receiving key.

He also adds a blinded **key tie**, labelled **T**. The note commits to T together with its ciphertexts. Later this ties the key used in the issuer proof to the key Carol actually holds. Without it, an honest recipient could be tricked into naming the wrong certified identity.



A2.2. Seal the tools needed to open and prove it

Carol needs more than ciphertexts. Her eventual proof needs the opening secret, value, issuer-encryption randomness, Bob's separate naming salt and the secret that opens T. Bob wraps each field with a distinct mask derived from the value envelope's shared encryption material.

That protects the construction's seams. Sending the issuer randomness or T's opening in the clear would reveal Bob through another route. Sending the naming salt openly would help test his membership. The outer package must protect its inner tools as carefully as its apparent message.

BOB

SEALS THE INNER MATERIAL

- Value + identity envelopes
- Binding + key tie T
- 🔒 Opening + naming material

MALLORY OUTSIDE OBSERVER

CAN COPY, NOT UNWRAP

- Ciphertexts + T
- 🔒 Each secret field masked
- ✗ No opening or naming salt

CAROL

CAN REBUILD THE WRAP KEY

- Receiving secret
- Public envelope component
- + **Shared wrap material**

Bob derives wrapping material from Carol's public key; Carol derives it locally

ADDRESSED DELIVERY

Wraps value, opening secret, issuer randomness, naming salt and tie-opening blind.
Each field has its own mask. None of these disclosure keys travels in the clear.

A2.3. The mint verifies Bob's claim and escrows the batch

Notes checks value conservation, insertion into the current note tree, the private-issuer mode and Bob's issuer-binding proof. The mint proof also ties the checked ciphertexts and T to the committed notes. Funding and insertion remain atomic.

Mallory sees Bob's minting *address*, 500 BUCK, commitments, encrypted issuer material and blinded proofs. A2 conceals the issuer's identity inside those objects; it does not conceal which address submitted them. Anyone who already knows that address belongs to Bob keeps that knowledge.

BOB

MINTS THE PRIVATE BATCH

- Sealed A2 delivery
- + **Mint proof + issuer proof**
- ✓ 500 BUCK escrowed

MALLORY OUTSIDE OBSERVER

READS PUBLIC FUNDING

- B → pool: 500 BUCK
- Commitments + envelopes
- Blinded proofs + T

CAROL

CAN CHECK THE PUBLIC ANCHOR

- Receiving secret
- + **Mint reference**
- Still needs delivery

Bob → Notes: commitments, encrypted issuer data, key ties and checked proofs

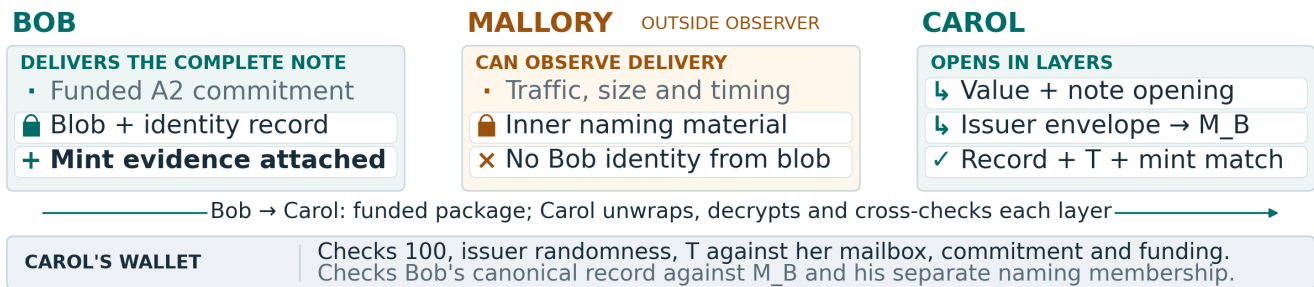
NOTES / BUCK

Checks A2 mode, value conservation, issuer binding and committed ciphertext ties.
Escrows the batch atomically. Minting address and total remain public.

A2.4. Carol unwraps, decrypts and cross-checks

Bob delivers the blob with the full mint evidence and his identity record over protected communication. Carol unwraps it with her receiving secret, checks the value and opening, and reconstructs the funded commitment.

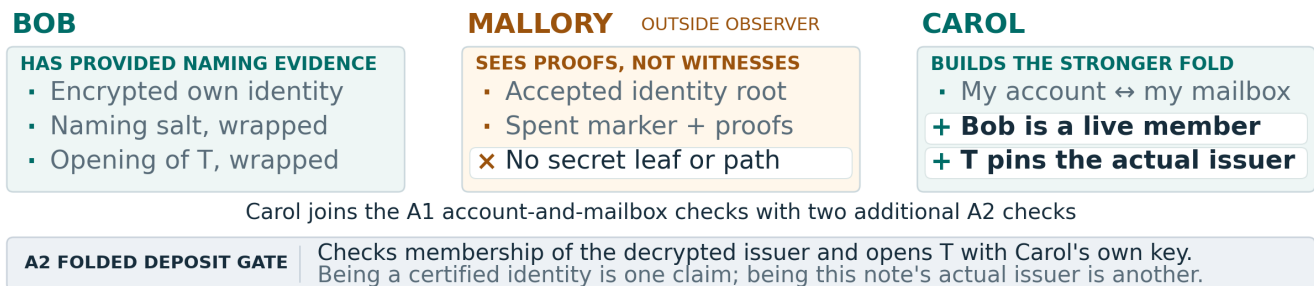
She decrypts Bob's point, compares his canonical record, and checks T against her actual receiving key. Bob's naming salt lets her locate and prove his separate certified naming association. It exposes that association to Carol; it does not give her Bob's receiving secret or his other salts. The distinction is deliberate and worth preserving in backups.



A2.5. Join the extra checks into the spend proof

Carol builds the addressed fold with her account identity, receiving secret and certified association. A2 adds two checks: the decrypted issuer is a member of the accepted identity tree, and T opens using *Carol's own* receiving key.

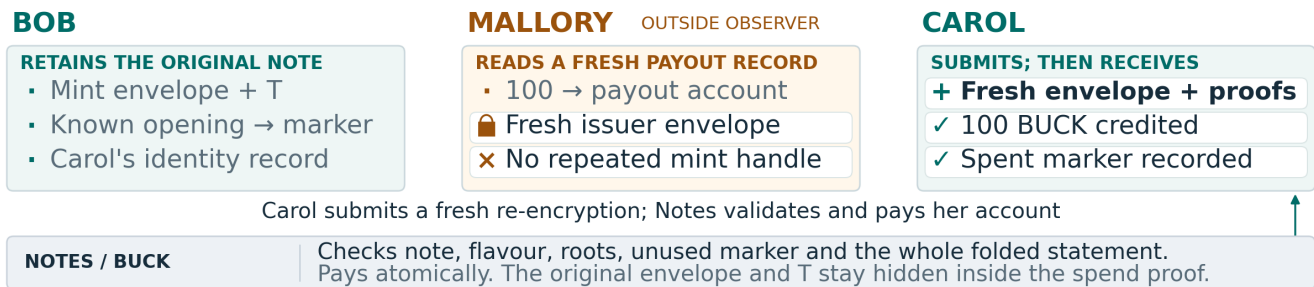
The first prevents a spendable note that names an uncertified invented point. The second prevents one that names a different certified person. "Someone certified" and "the person who wrote this note" are different claims. The proof needs both, and binds them to the same note marker.



A2.6. Rewrap for the public spend; then execute

Carol produces a *fresh* encryption of the same issuer identity for her mailbox. Her proof checks its relationship to the original note. Publishing the original mint envelope again would hand Mallory an exact match.

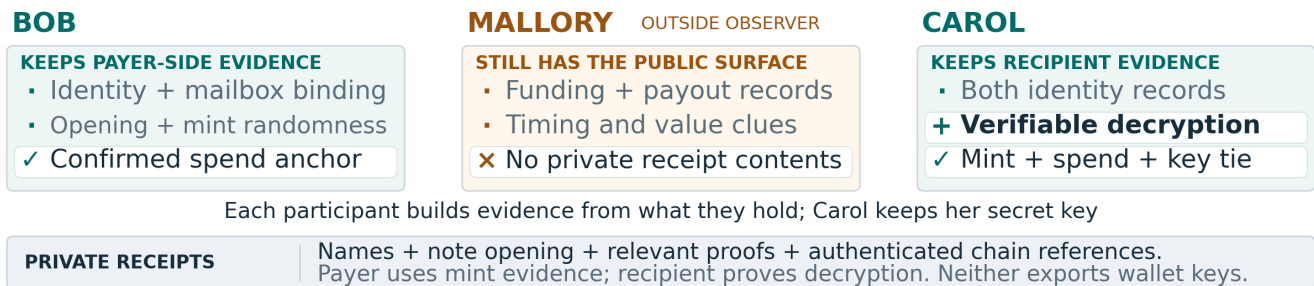
Notes verifies the funded note, unused marker, correct style, accepted roots and complete folded statement, then pays Carol atomically. The original envelope, T and private paths remain inside the proof's hidden inputs. Mallory sees 100 BUCK leave for an address, together with fresh ciphertext and proofs. She still has timing and amount clues, not a newly published exact mint handle.



A2.7. Keep a useful receipt

Both parties retain their identity records, note opening, relevant proofs and confirmed chain references. Bob's evidence comes from his mint randomness and Carol's certified mailbox association. Carol can prove what she decrypted without disclosing her receiving secret; her receipt also checks the issuer binding and T.

A receipt reader learns the disclosed names and this payment's links. They do not thereby receive either citizen's credential or wallet keys. The current full-record receipt carries more identity data than its short printed label suggests; encoding it as a blob is not encryption. Keep it as deliberately as any other private financial record.⁹

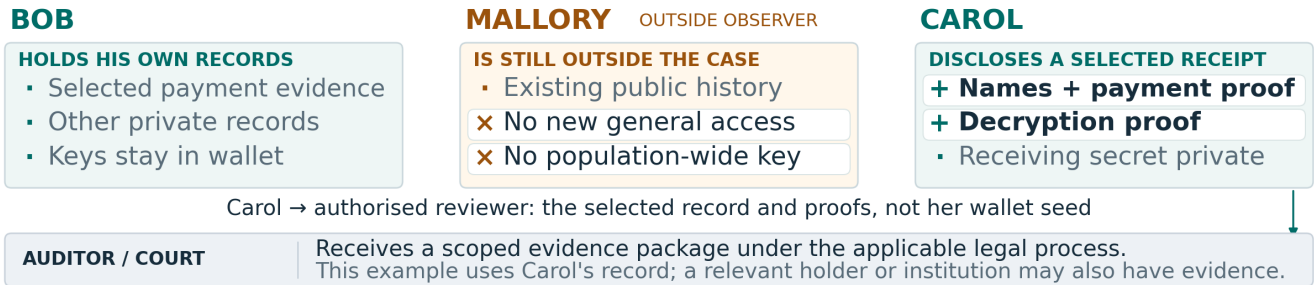


Private commerce, usable evidence

R1. Disclose the relevant record

A dispute can start with an ordinary citizen producing a receipt. An auditor or investigator acting under the applicable legal process can seek relevant records from a participant or another holder of the evidence. Here Carol supplies a selected receipt and decryption proof. She need not give away her seed or receiving secret to establish what that envelope said.

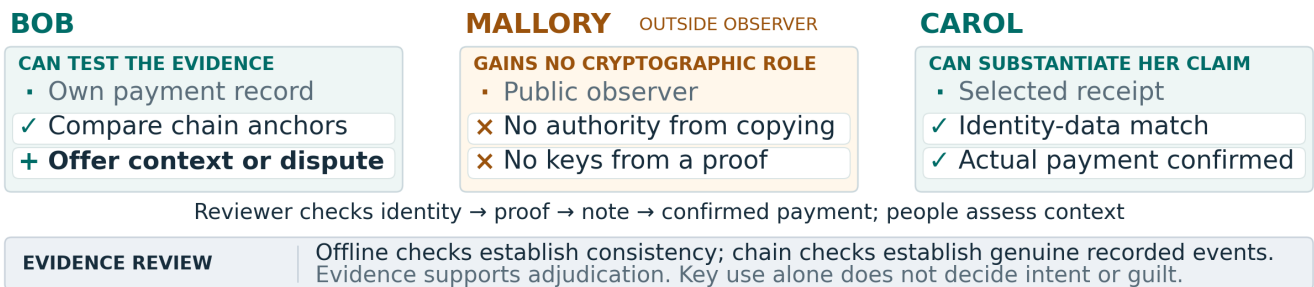
This is a design for evidence held by people who actually took part. Institutions also hold their direct relationships, as N1 showed. Privacy does not depend on pretending those records do not exist.



R2. Check the record; let people judge the case

The reviewer first checks internal consistency: names match identity fingerprints, proofs fit their envelopes, and the note opening matches the claimed payment. Then authenticated chain records establish that the registration, mint and successful spend actually occurred. A plausible offline packet alone does not prove a payment happened.

Bob can examine the same evidence and offer context or contest its use. Certification quality, compromised keys and human intent remain questions for investigation and adjudication. A signature is evidence of key use, not a verdict. Good cryptography gives civil governance better facts to work with while leaving citizens room to conduct ordinary private commerce.



What Mallory can join

The practical gain is choice over relationships, not an invisible ledger. This table concerns a stranger with chain access and no participant secrets. Public identity records and outside knowledge can add to her view.

Payment	Public connection	Identity disclosure	Who can recognise the note?
Direct EOA	Exact sending and receiving addresses; amount	Private names in bilateral envelopes	Participants; anyone with an address-to-name link can attach it
B1	Exact mint commitment at redemption; public issuer; payout and amount	Issuer public; depositor encrypted for issuer	Public observers link mint and spend; opening holders can also recognise it
A1	Public mint and payout records; no exact mint handle at spend	Issuer public; recipient privately bound	Opening holders; outsiders retain timing and value clues
A2	Public mint and payout records; fresh ciphertext at spend	Both identities concealed from an otherwise uninformed outsider	Opening holders; outsiders retain timing and value clues

An individual can use A2 to move value between accounts of their own identity. It removes the direct address-to-address transfer edge that an ordinary transfer would publish. Reusing an address still links activity on that address. A distinctive amount, a sparse pool, immediate redemption, common gas funding or observable delivery can suggest links across accounts. Cryptographic independence is useful; operational independence must accompany it.

Five boundaries make the positive claim precise:

- **Counterparties know their own dealings.** A note issuer retains its openings and can recognise their eventual spends. Disclosure of a receipt shares some of those links. It is not a general view into the other citizen's accounts.
- **Certification is a public responsibility.** An issuer knows its clients and can issue credentials; a dishonest issuer can falsely certify or duplicate an identity. Hiding presentations limit recognition of honest holders' registrations, not the issuer's power to lie.
- **Current standing is a governed fact.** Membership authorities can remove an association; accepted older roots provide a bounded transition window. Notes currently accepts retained roots up to seven days old. This affects eligibility without publishing all the holder's accounts. Ordinary EOA transfer checks do not themselves impose this fresh-membership test.
- **Intermediaries have a view.** Pool operators, custodial wallets and data services can accumulate their direct observations. Their security and retention practices matter. Notes does not promise that one compromised institution can never disclose many customers.
- **Secrets require recovery and care.** The seed restores the receiving key; the identity record does not. A leaked long-lived receiving secret exposes addressed material available under that key, including old deliveries. A new account alone does not replace it. Complete wallet compromise and future breaks of the underlying cryptography remain outside these promises.

The useful arrangement is modest: let strangers verify that the rules were followed, let citizens recognise and document their own dealings, and let public institutions examine evidence through accountable processes. Buying a bicycle should not require publishing the rest of your life.

Sources and reading notes

This account was checked against checkout 2183f41 on 2026-09-24. It follows the review reading plan, including the receiving-key architecture and accumulator specification, rather than treating older summaries as current security guarantees. The full source trail is deliberately kept here so the citizen's explanation need not carry the algebra.

Notes

¹Proofs: review status; Review index; Receiving-key reading plan. The later implementation notes supersede some earlier status paragraphs. The underlying signature reference is Pointcheval and Sanders, Short Randomizable Signatures. It is not an independent validation of this project’s masked presentation or composition.

²Identity, Identity Example, and Accumulator specification, especially the distinct receiving and mailbox leaves. The wallet derives the receiving secret and salts from its seed in `recvkey.py` and `salt.py`. The identity record is frozen at first certification; mutable standing and attributes are separate memberships. The diagrams specify the required certified association and wallet-only key custody. They do not invent an enrollment wire protocol: the secret-preserving issuer-side certification of the hidden spending commitment still needs a fully specified and validated ceremony. The local tree insertion helpers are not themselves that ceremony.

³Ethereum’s official descriptions of accounts and transactions distinguish key-controlled EOAs from code-controlled contract accounts and describe signed transaction authorisation. BUCK’s identity and Notes keys are additional application-level keys.

⁴Buck.sol, the two `approve` overloads and `_identityCheckedTransfer`; IdentityRegistry.sol, `_verifyCredential` and `_verifyApprove`. The identity proof binds the registered keys and identity relationship; the Ethereum transaction signs the actual call and allowance. A verified proof is not a grant of unlimited spending authority. Failed transactions can still expose their submitted data and consume transaction fees.

⁵Notes: the pool account, and Buck.sol’s identity guards. Public-identity endpoints can use the public fallback; private endpoints must supply their directional identity approval. The pool operator’s approval-decryption capability is distinct from Carol’s independent receiving secret.

⁶IdentityRegistry.sol, `bindContract`, target authorisation and `_storeBinding`; Notes.sol, public `mint` overload and `verifyIssuerSchnorr`. This example uses a controlled public contract, not an unimplemented “make this EOA public” operation.

⁷Notes.sol, `spendCoupledB1`, checks `publicIssuerOfCommitment[issuanceCommitment]`. The explicit commitment is also an input to the spend proof. Broad “no mint-to-spend link in any flavour” passages in the Notes papers do not describe this B1 interface. Serialized Notes was reviewed as a separate research proposal; its family-linked sub-notes are not used here. The editorial review records unresolved conservation and disclosure issues in that proposal.

⁸Notes Flow: current spend paths, Receiving-key architecture, `delivery.py`, A1 folded gate and A2 folded gate. The current A1 fold binds the committed *value* ciphertext, not a claim that its auxiliary delivery identity ciphertext is independently part of `idHash`. Addressed folds authenticate `msg.sender`; the note-spend proof binds the chosen payout address. Here Carol submits and receives into the same account. The contract also permits an authenticated holder to choose a different payout, subject to BUCK’s endpoint rules; it does not prove the submitter owns every possible payout account. B1 instead binds the depositor identity to its payout account.

⁹Receipts, particularly the point-to-name bridge, evidence-bearing addressed receipt legs and the two verification tiers; `receipt.py`. A full identity record discloses its derived identity scalar to its reader. The receiving secret was separated from that scalar precisely to prevent a receipt from becoming a mailbox key. Availability of canonical identity data remains a separate obligation from the chain’s identity-point proofs.

Illustration sources

The 32 narrow illustrations are generated from the figure descriptions by `render_privacy_figures.py`. Rebuild the PNGs and editable SVGs in `images/privacy/` with:

```
python3 scripts/render_privacy_figures.py
```

Org and PDF use the PNGs at full text width; the text and symbols remain legible without colour. Each illustration has an alternative description and the surrounding step explains its contents.

Render the paper with `make doc-alberta-buck-privacy` in the normal Emacs/L^AT_EX environment. This document contains no executable protocol transcript and does not regenerate circuit artifacts.